

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	6ab4ce760fc666a751b52130
Project Name	AI StudyBuddy
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	2-space indentation used consistently across all files
2	Proper File Structure	Files and folders are logically organized	Yes	MVC structure: src/controllers, middleware, models, routes, utils
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g., accessToken, refreshToken, materialId, askGemini
4	Function / Method Names	Functions are descriptively named	Yes	e.g., register, login, uploadMaterial, generateFlashcards, summarize
5	Code Comments	Inline and block comments explain logic	Yes	Route comments (e.g., // POST /api/auth/register), inline explanations
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate controllers, middleware, models, routes, and utility modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Shared utility functions (askGemini, generateTokens) prevent duplication
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Global error handler middleware + express-async-errors for automatic catching

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	askGemini(prompt) — utils/gemini.js	JavaScript / @google/generative-ai	Used in materialController (summarize, flashcards, quiz, study-plan) and aiController (flashcards, quiz, study-plan)	High
2	generateTokens(userId, role) — utils/tokens.js	JavaScript / jsonwebtoken	Used in authController (register, login, refresh)	High
3	protect() middleware — middleware/auth.js	JavaScript / jsonwebtoken	Used in all protected route files (materials, ai, admin)	High
4	adminOnly() middleware — middleware/auth.js	JavaScript / Express	Used in admin routes for RBAC enforcement	High
5	upload middleware — middleware/upload.js	JavaScript / Multer	Used in materials routes for file upload handling	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean MVC folder structure with logical separation of concerns
Readability	5	Descriptive naming, consistent formatting, route comments on every handler
Reusability	5	Shared utilities (askGemini, generateTokens), reusable middleware (protect, adminOnly)
Documentation / Comments	4	Route-level comments present; inline comments for key logic; API documentation provided
Overall Score	4.75	Well-structured, maintainable, and production-ready codebase